

VIVEK KUMAR MISHRA

EB-1A Petition for Alien of Extraordinary Ability

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PROFESSIONAL SUMMARY

IEEE Senior Member and Sr. Software Engineer with extraordinary ability in Applied Artificial Intelligence and Machine Learning, demonstrated through federal government archival of research methodologies (U.S. National Transportation Library), adoption by state agencies (Florida Department of Transportation), and critical role at systemically important financial institution (JPMorgan Chase) developing production AI systems serving 80+ million users. Proven expertise validated through peer review service at premier international conferences (NeurIPS 2025, ICLR 2025) and formal curriculum adoption at government-regulated academic institutions.

EDUCATION

Master of Science in Computer Science

California State University, Long Beach (CSULB) – Long Beach, California
Graduated: August 2021 | GPA: 3.8/4.0

Bachelor of Science in Information Technology

West Bengal University of Technology (now Maulana Abul Kalam Azad University of Technology)
Graduated: May 2014

PROFESSIONAL EXPERIENCE

Senior Software Engineer

October 2021 – Present

JPMorgan Chase & Co. – Wilmington, Delaware

- Served as **founding engineer** on the Chase Shopping Hub platform, designing and implementing backend APIs and microservices architecture for e-commerce platform serving **80+ million Chase cardmembers**
- Architected scalable RESTful APIs using Java, Spring Boot, and AWS (DynamoDB, SQS, ECS, S3) processing **5+ million customer requests monthly** with **99.9% reliability**
- Reduced operational costs by **\$100,000 annually** through cloud-based resource optimization and automated cost-control solutions
- Improved application performance by **43%** through backend microservices optimization and distributed systems design improvements
- Led team of 3 developers in deploying six production-grade APIs supporting high-availability requirements for mission-critical financial services

- Implemented comprehensive CI/CD pipeline using Jenkins, Docker, Bitbucket, and Spinnaker, improving deployment efficiency and developer experience
- Increased automated test coverage from **17% to 97%** using JUnit and Mockito, significantly improving code quality and system reliability
- Built observability infrastructure using Datadog, Splunk, and Dynatrace for real-time monitoring of distributed systems
- Integrated OAuth 2.0, OIDC, SAML, and JWT for secure API authentication serving millions of customers

Research Assistant

March 2020 – August 2021

California State University, Long Beach – Department of Civil Engineering – Long Beach, California

- Recruited by Civil Engineering department for specialized Python and machine learning expertise for **federally-funded intelligent transportation systems research** (U.S. Department of Transportation via Mineta Transportation Institute)
- Developed machine learning methodologies for transportation data analysis using NumPy, Pandas, TensorFlow, and advanced NLP techniques
- Created Python frameworks for keyword indexing and text data processing for transportation research applications
- Authored research papers resulting in **permanent archival in U.S. National Transportation Library** (ROSA P Records dot/61210, dot/59030)
- Research methodologies subsequently **adopted by Florida Department of Transportation** in federally-funded infrastructure programs

Member of Technical Staff

July 2018 – January 2020

Oracle Financial Services Software Limited – Bangalore, India

- Automated Oracle Cloud Infrastructure (OCI) testing workflows using Ansible, Python, and Go for storage, compute, and networking modules
- Developed and deployed infrastructure-as-code solutions using Terraform and Docker for distributed cloud systems
- Migrated legacy Python 2.x codebases to Python 3.x and deployed to OCI using continuous integration pipelines
- Commanded salary **83% above market average** (11,00,000 vs. 6,00,000 market average) based on exceptional technical capabilities

Programmer Analyst

August 2014 – July 2018

Cognizant Technology Solutions – Bangalore, India

- Automated 500+ test cases using Python and shell scripting, establishing comprehensive regression testing framework

- Reduced continuous integration runtime from 4.5 hours to 2.5 hours, saving **500 hours annually**
- Configured databases and backend applications for enterprise-scale financial services systems
- Earned **Global Project of the Year Award** for outstanding technical contributions

RESEARCH PUBLICATIONS (Peer-Reviewed)

1. **V. Mishra** et al., “Integrating Solar Photovoltaic Systems into the Grid: An Overview of AI Application,” *IEEE Access* (Q1 journal, Impact Factor 3.9)
2. **V. Mishra** et al., “The Role of Generative AI in Revolutionizing Healthcare, Education, and Finance: A Mini Review,” *Electronics* (Q2 journal)
3. **V. Mishra** et al., “Framework for Rail Transport Inequality Assessment,” *MDPI Sustainability* (Impact Factor 3.9, Scopus Q2) – **15 citations**
4. **V. Mishra** et al., “Evaluating Financing Mechanisms and Economic Benefits to Fund Grade Separation Projects” – **12 citations**
5. **V. Mishra** et al., “Evaluating Innovative Financing Mechanisms for the California High-Speed Rail Project” – **7 citations**
6. **V. Mishra** et al., “Funding the High-Speed Rail: A Case Study of the California Project” – **3 citations**
7. **V. Mishra** et al., “Optimizing Multimodal Transportation Access to Support Commuting Among Low-Income Transit Riders”
8. Additional publications archived in **U.S. National Transportation Library** (ROSA P Records dot/61210, dot/59030)

Total Citations: 37 | **h-index:** 3 (Google Scholar)

TECHNICAL BOOK

V. Mishra, D. Panwar, R. Vishwakarma, “Design Patterns for AI Agents: A Structured Approach to Building Robust and Intelligent Systems,” ISBN: 979-8268912371, 2024

- Comprehensive 300+ page technical framework for AI agent architecture
- **Formally adopted** by Anand International College of Engineering (AICTE-approved) for permanent library collection
- **Integrated into curriculum** at Supaul College of Engineering (Government of Bihar) for 47 students in CSE(AI) program
- Recognized by industry practitioners as practical engineering reference

PROFESSIONAL SERVICE & PEER REVIEW

Premier Conference Reviewer

- **NeurIPS 2025 Position Paper Track Reviewer** – Conference on Neural Information Processing Systems (24.52% acceptance rate, H5-index 278)
- **ICLR 2025 FM-Wild Workshop Reviewer** – International Conference on Learning Representations (32% acceptance rate, H5-index 362)
- Graph Signal Processing Workshop 2025 Reviewer
- ACDSA 2026 Reviewer (International Conference on AI, Computer, Data Sciences and Applications)

IEEE Senior Member Application Review Panel – June 2025

Evaluated professional applications for IEEE Senior Member elevation, assessing candidates’ demonstrated “significant performance” over five years

Journal Peer Reviewer

- Informatica (active reviewer with multiple assignments)
- Virtus InterPress: Risk Governance and Control journal
- Virtus InterPress: Corporate & Business Strategy Review

Hackathon Judge

- PennApps (University of Pennsylvania) – Certified Judge
- DragonHacks 2024
- Claro 2024

PROFESSIONAL MEMBERSHIPS

IEEE Senior Member (Member #100495418) – November 2024

Elevated to Senior Member grade based on outstanding achievements. Senior Member status requires:

- Ten years of professional practice
- Five years of significant performance (evaluated by three IEEE Senior Members/Fellows)
- Only 8% of IEEE’s 460,000+ members achieve this distinction

IEEE Signal Processing Society – Active Member

IEEE Young Professionals – Active Member

MEDIA RECOGNITION & EXPERT QUOTATIONS

IEEE Official Recognition

- Featured as “IEEE Young Creator” on IEEE Transmitter (460,000+ IEEE members)
- Official speaker profile on IEEE.tv (award-winning IEEE television network)
- Featured twice on IEEE’s official Twitter/X account (342,900 followers)

Major Press Coverage

- **Dainik Bhaskar** (India’s #1 Hindi newspaper, 63 million readers) – December 2024
- **Rajasthan Patrika** (25.9 million readers) – Keynote presentation coverage
- **First India News** (4.4M+ YouTube subscribers)

Expert Source in Technical Trade Publications

- **TechBullion** – Exclusive interview on backend engineering and cloud innovation (230K monthly visitors)
- **TechTarget** – Quoted in “The Future of Generative AI: 10 Trends to Follow in 2025” (1.4M+ daily readers)
- **TechTarget** – Quoted in “Generative AI’s Sustainability Problems Explained”
- **Dice.com** – Quoted as IEEE Senior Member expert in “Database Administrators and AI” (2.4M monthly visitors)
- **VKTR.com** – Quoted in “Do AI Coding Tools Really Increase Developer Productivity?” (7M+ readers)

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, Go, Shell Scripting

Frameworks: Spring Boot, Spring Security, React, TensorFlow, NumPy, Pandas

Cloud & Infrastructure: AWS (VPC, DynamoDB, IAM, SQS, ECS, S3), Terraform, Docker, Kubernetes, Ansible

CI/CD & DevOps: Jenkins, Spinnaker, Bitbucket, Jules

Monitoring: Datadog, Splunk, Dynatrace, Grafana

Messaging: Kafka (event-driven architecture)

Security: OAuth 2.0, OIDC, SAML, JWT, Spring Security

AI/ML Tools: LLMs, GenAI, OpenCV, NLP, Cursor, Windsurf

Testing: JUnit, Mockito, JMeter

CERTIFICATIONS

AWS Certified Cloud Practitioner – Amazon Web Services

HONORS & AWARDS

- **Global Recognition Award 2024** – JPMorgan Chase
- **#FutureOfMeterReading Contest Winner** – Capgemini Super Techies
- **Global Project of the Year Award** – Cognizant Technology Solutions (2x recipient)

GOVERNMENT VALIDATION & INSTITUTIONAL ADOPTION

- **U.S. Department of Transportation:** Research methodologies permanently archived in National Transportation Library (official federal repository)
- **Florida Department of Transportation:** Machine learning methodologies adopted in federally-funded infrastructure research programs (cross-state validation)
- **Supaul College of Engineering** (Government of Bihar): Technical book formally integrated into Computer Science & AI curriculum as required reference for 47 students
- **Anand International College of Engineering** (AICTE-approved): Technical book adopted into permanent Central Library collection

This resume is submitted as part of an I-140 petition for EB-1A classification
(Alien of Extraordinary Ability in Applied Artificial Intelligence and Machine Learning)